

DIPOLE TYPE BY LENGTH  
*l* = antenna rod length (m)

| $\lambda$ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m) |                  |                                                           |
|----------------------------------------------------------------------|------------------|-----------------------------------------------------------|
| <i>l</i> /λ                                                          | Type             | Use formula                                               |
| < 1/50                                                               | Hertzian (short) | HERTZIAN $S(R, \theta)$ , $D=1.5$                         |
| = 1/4                                                                | Quarter-wave     | ARB formula, $\pi l/\lambda = \pi/4$                      |
| = 1/2                                                                | Half-wave        | half-wave shortcut, $D=1.64$ , $R_{\text{rad}}=73 \Omega$ |
| = 1                                                                  | Full-wave        | ARB, $D=2.41$ , $R_{\text{rad}}=199 \Omega$               |
| other                                                                | Arbitrary        | ARB formula                                               |

HERTZIAN DIPOLE —  $S(R, \theta)$

|                                                                                                                                                                                                                                                |                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Far-field E,H phasors <small><i>small l</i> ≪ λ</small><br>$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} \frac{e^{-jkR}}{R} \sin \theta$ , $\vec{H}_\phi = \vec{E}_\theta / \eta_0$<br>Power density formulas <i>l</i> /λ < 1/50; far field |                                                                                 |
| Power density                                                                                                                                                                                                                                  | $S(R, \theta) = S_{\text{max}} \sin^2 \theta$ (W/m <sup>2</sup> )               |
| Max pwr density                                                                                                                                                                                                                                | $S_{\text{max}} = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}$ (W/m <sup>2</sup> ) |
|                                                                                                                                                                                                                                                |                                                                                 |
| <i>l</i> : rod length (m)                                                                                                                                                                                                                      | $I_0$ : peak current (A)                                                        |
| <i>R</i> : obs. distance (m)                                                                                                                                                                                                                   | $\theta$ : angle from dipole axis (rad)                                         |
| <i>k</i> =2π/λ (rad/m)                                                                                                                                                                                                                         | $\eta_0=120\pi \approx 377 \Omega$                                              |
| Total radiated power                                                                                                                                                                                                                           | $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)                   |

ARBITRARY-LENGTH DIPOLE —  $S(\theta)$

|                                                                                                                                                                                                                                                                                                                                                                |                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Power density at <i>R</i> any <i>l</i><br>$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[ \frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$<br><i>At θ = 0, π: bracket is 0/0. L'Hôpital ⇒ S = 0 (no axial radiation). TI-36X Pro alt: evaluate bracket at θ = 0.001 rad → tiny → 0.</i> |                                          |
| Normalized pattern <small>dimensionless</small>                                                                                                                                                                                                                                                                                                                | $F(\theta) = S(\theta) / S_{\text{max}}$ |
| COMMON ANGLES (deg ↔ rad, trig values)                                                                                                                                                                                                                                                                                                                         |                                          |

Conversion *multiply by  $\pi/180$  or  $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$$

| °   | rad     | sin $\theta$ | cos $\theta$ | sin <sup>2</sup> $\theta$ | cos <sup>2</sup> $\theta$ |
|-----|---------|--------------|--------------|---------------------------|---------------------------|
| 0   | 0       | 0            | 1            | 0                         | 1                         |
| 30  | $\pi/6$ | 1/2          | $\sqrt{3}/2$ | 1/4                       | 3/4                       |
| 45  | $\pi/4$ | $\sqrt{2}/2$ | $\sqrt{2}/2$ | 1/2                       | 1/2                       |
| 60  | $\pi/3$ | $\sqrt{3}/2$ | 1/2          | 3/4                       | 1/4                       |
| 90  | $\pi/2$ | 1            | 0            | 1                         | 0                         |
| 180 | $\pi$   | 0            | −1           | 0                         | 1                         |

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

|                                                                                                                                                                                               |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <i>N</i> identical elements along axis, spacing <i>d</i> , equal phase (broadside, $\theta_{\text{max}} = \pi/2$ ). Total pattern $S = S_e \cdot F_a$ ( $S_e$ = single elem).                 |  |
| Array factor (uniform, broadside) $k = 2\pi/\lambda$ , $\gamma = kd \cos \theta$<br>$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$ ; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$ |  |
| Normalize $F_a^{\text{norm}} = F_a / N^2$                                                                                                                                                     |  |
| HPBW (broadside, uniform) <small>use <i>Nd</i>, not (<i>N</i>−1)<i>d</i></small><br>$\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{Nd}$                     |  |
| <i>Grating lobes appear if d &gt; λ. Avoid.</i>                                                                                                                                               |  |

BEAM PARAMETERS — β, Ω<sub>p</sub>, *D*

|                                                                                                                                                                                                                                                                                                                                                                                                     |                                     |                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------|
| Normalize <small>always start here</small><br>$F(\theta) = S(\theta) / S_{\text{max}}$ (unitless)<br>Beamwidth β at threshold <i>X</i><br>Set $F(\theta_X) = X$ , solve for θ <sub>X</sub> ; β = 2θ <sub>X</sub> (symmetric)                                                                                                                                                                        |                                     |                                   |
| Threshold                                                                                                                                                                                                                                                                                                                                                                                           | $X = 10^{\text{dB}/10}$             | θ <sub>X</sub> (Gauss. short-cut) |
|                                                                                                                                                                                                                                                                                                                                                                                                     |                                     |                                   |
| HPBW (−3 dB)                                                                                                                                                                                                                                                                                                                                                                                        | 0.5                                 | $\sqrt{\frac{\ln 2}{a}}$          |
| −6 dB                                                                                                                                                                                                                                                                                                                                                                                               | 0.25                                | $\sqrt{\frac{\ln 4}{a}}$          |
| −10 dB                                                                                                                                                                                                                                                                                                                                                                                              | 0.1                                 | $\sqrt{\frac{\ln 10}{a}}$         |
| any $F(\theta_X)$                                                                                                                                                                                                                                                                                                                                                                                   | any <i>X</i>                        | $F(\theta_X) = X$                 |
| Pattern solid angle Ω <sub>p</sub> <small>recognize pattern, look up</small><br>$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr)<br><i>No φ dep.</i> : Ω <sub>p</sub> = 2π ∫ <sub>0</sub> <sup>π</sup> <i>F</i> (θ) sin θ <i>dθ</i> .<br><i>θ is the integration variable — DON'T plug in a number for θ (e.g. θ<sub>1/2</sub>). Integrate over θ ∈ [0, π].</i> |                                     |                                   |
| Pattern <i>F</i> (θ)                                                                                                                                                                                                                                                                                                                                                                                | Ω <sub>p</sub> (sr)                 |                                   |
| Isotropic ( <i>F</i> =1)                                                                                                                                                                                                                                                                                                                                                                            | 4π ≈ 12.57                          |                                   |
| Hertzian sin <sup>2</sup> θ                                                                                                                                                                                                                                                                                                                                                                         | 8π/3 ≈ 8.38                         |                                   |
| Half-wave dipole                                                                                                                                                                                                                                                                                                                                                                                    | ≈ 7.66                              |                                   |
| Narrow Gauss $e^{-a\theta^2}$                                                                                                                                                                                                                                                                                                                                                                       | $\pi/a = \pi\theta_{1/2}^2 / \ln 2$ |                                   |
| 2-axis HPBW (aperture)                                                                                                                                                                                                                                                                                                                                                                              | β <sub>xz</sub> · β <sub>yz</sub>   |                                   |
| Other (use integral)                                                                                                                                                                                                                                                                                                                                                                                | numerical                           |                                   |

TI-36X Pro: [2nd] [d/dx] for  $\int_{\square}^{\square}$ , RAD mode, multiply by 2π.

|                                                                                                                                                                                                                                                                                                                                                                                             |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Directivity <i>D</i> <small>always <math>D = 4\pi/\Omega_p</math>; common shortcuts below</small><br>$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$                                                                                                                                                                                                                |  |
| WORD TRAP: “direction” = θ <sub>max</sub> (rad); “directivity” = <i>D</i> (unitless).<br>Dipoles → COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p / \lambda^2$ . 2-axis HPBW: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$ . Circular: $D \approx 4\pi / \beta^2$ .<br>Gain (if asked) $\xi = \text{rad. eff.}$ , $0 < \xi \leq 1$<br>$G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$ |  |
| Lossless / ideal antenna → $G = D$ .                                                                                                                                                                                                                                                                                                                                                        |  |

COMMON DIPOLE PARAMETERS

| Lossless (ξ=1): <i>G</i> = <i>D</i> . |           |          |                        |                                              |
|---------------------------------------|-----------|----------|------------------------|----------------------------------------------|
| Type                                  | <i>l</i>  | <i>D</i> | <i>D</i> <sub>dB</sub> | <i>R</i> <sub>rad</sub> (Ω)                  |
| Isotropic                             | N/A       | 1        | 0                      | —                                            |
| Hertzian                              | < λ/50    | 1.5      | 1.76                   | 80π <sup>2</sup> ( <i>l</i> /λ) <sup>2</sup> |
| Half-wave                             | λ/2       | 1.64     | 2.15                   | 73.2                                         |
| Monopole                              | λ/4 (gnd) | 1.64     | 2.15                   | 36.6                                         |
| Full-wave                             | λ         | 2.41     | 3.82                   | 199                                          |

Half-wave:  $P_{\text{rad}} = 36.6 I_0^2$  W. Monopole (λ/4 above ground): same patt. as half-wave but  $P_{\text{rad}}$ ,  $R_{\text{rad}}$  halved.

ANTENNA AREAS (*A<sub>e</sub>*, *A<sub>p</sub>*)

|                                                                                                                                                                                                                                           |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Effective area <small>antenna's RX cross section; <i>D</i> from BEAM PARAMS or table above</small><br>$A_e = \frac{\lambda^2 D}{4\pi}$ (m <sup>2</sup> )                                                                                  |  |
| Physical cross section <small>wire dipole, diameter <i>d</i>; <i>l</i> see table above</small><br>$A_p = l d$ (m <sup>2</sup> , any dipole)<br>Inverse: <i>D</i> from <i>A<sub>e</sub></i><br>$D = \frac{4\pi A_e}{\lambda^2}$ (unitless) |  |
| Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$ ).                                                                                                                                                |  |

FRISIS TRANSMISSION FORMULA

For each antenna's *G*: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (*G* = *D*, lossless). Else (dB/linear value given) → use as given (convert dB → linear).

Pwr density at RX, then RX pwr  $G_t, G_r$  linear gains;  $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } \left[ P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t \right] \text{ (W)}$$

Equivalent:  $P_r = S_r A_{e,r}$ .

Solve for *R* max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$$

Equivalent form using *A<sub>e</sub>* recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss:  $L_{fs} = (4\pi R/\lambda)^2$ , so  $P_r = G_t G_r P_t / L_{fs}$ .

Off-axis (patterns not aligned): multiply by  $F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$ , both = 1 when aligned.

Recipe 1. λ = *c* / *f*. 2. Convert any dB gain → linear:  $G = 10^{G_{\text{dB}}/10}$ .

3. Get *G<sub>t</sub>*, *G<sub>r</sub>* by antenna type: dipole→COMMON DIPOLE PARAMS; aperture→APERTURE box; arbitrary *F* →BEAM PARAMS. 4. Plug into Friis.

Use linear *G*, not dB. Convert FIRST.

dB ↔ LINEAR

Works for any power-like quantity (*G*, *D*, *P<sub>r</sub>*/*P<sub>t</sub>*, *SNR*, ...):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

| dB | linear | dB  | linear |
|----|--------|-----|--------|
| 0  | 1      | 10  | 10     |
| 3  | 2      | 13  | 20     |
| 6  | 4      | 20  | 100    |
| 7  | 5      | 30  | 1000   |
| −3 | 0.5    | −10 | 0.1    |

Memorize: 3 dB ↔ ×2, 10 dB ↔ ×10. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power ∝ *E*<sup>2</sup>).

NOISE & SNR

|                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Noise power <small>thermal noise into matched receiver</small><br>$P_N = k_B T_{\text{sys}} B$ (W)<br>$k_B = 1.38 \times 10^{-23}$ J/K; <i>T<sub>sys</sub></i> system noise temp (K); <i>B</i> receiver bandwidth (Hz).<br>Signal-to-noise ratio<br>$\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$<br>Typical comm. link needs SNR > 10 dB for usable signal. |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

APERTURE ANTENNAS (horn, dish)

*A<sub>p</sub>* physical aperture area (m<sup>2</sup>); *A<sub>e</sub>* effective area (m<sup>2</sup>, = λ<sup>2</sup>*D*/4π); β<sub>xz</sub>, β<sub>yz</sub> HPBW's in two planes (rad); *ℓ*, *d* aperture side / diameter (m).

|                                                                                                                                                         |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Directivity (any shape, uniform illum.)<br>$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) ⇒ $A_e \approx A_p$                                        |  |
| Directivity from beamwidths <small>any aperture</small><br>$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$ |  |

| HPBW and <i>D</i> by aperture shape <small>uniform illum.; β in rad</small> |                      |                                                                              |                                                                            |
|-----------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Shape                                                                       | <i>A<sub>p</sub></i> | HPBW (rad)                                                                   | <i>D</i> (unitless)                                                        |
| Rect. $\ell_x \times \ell_y$                                                | $\ell_x \ell_y$      | $\beta_x \approx 0.88\lambda/\ell_x$<br>$\beta_y \approx 0.88\lambda/\ell_y$ | $D \approx 4\pi / (\beta_x \beta_y)$<br>$= 4\pi \ell_x \ell_y / \lambda^2$ |
| Square $\ell$                                                               | $\ell^2$             | $\beta \approx 0.88\lambda/\ell$                                             | $D \approx 4\pi/\beta^2 = 4\pi \ell^2/\lambda^2$                           |
| Circular (diam. <i>d</i> )                                                  | $\pi d^2/4$          | $\beta \approx 1.02\lambda/d$                                                | $D \approx 4\pi/\beta^2 = \pi^2 d^2/\lambda^2$                             |

|                                                                                                                                                                                                                                                                                                                                                                                      |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Scaling (any ratio) $D \propto A_p/\lambda^2$ ; $\beta \propto \lambda/\sqrt{A_p}$<br>$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left( \frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$<br>$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$ |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

If a parameter is unchanged, its ratio = 1. Examples: double area → *D* ×2, β ×1/√2; double freq → *D* ×4, β ×1/2.

DIPOLE

|                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>l</i> — antenna length (m)<br>λ = <i>c</i> / <i>f</i> (m); <i>c</i> = 3 × 10 <sup>8</sup> m/s<br><i>I</i> <sub>0</sub> — peak current (A)<br><i>k</i> = 2π/λ (rad/m)<br><i>R</i> — obs. dist. (m); θ from axis<br>η <sub>0</sub> = 120π ≈ 377 Ω<br><i>S</i> ( <i>R</i> , θ) — pwr density (W/m <sup>2</sup> ) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

BEAM

|                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>F</i> (θ) = <i>S</i> / <i>S</i> <sub>max</sub> (unitless)<br><i>a</i> — coef. on θ <sup>2</sup> in Gaussian<br>β — HPBW (rad or °)<br>Ω <sub>p</sub> — pattern solid angle (sr)<br><i>G</i> — directivity<br>ξ rad. eff.; <i>G</i> = ξ <i>D</i><br><i>D</i> <sub>dB</sub> = 10 log <sub>10</sub> <i>D</i> ; <i>D</i> = 10 <sup><i>D</i><sub>dB</sub>/10</sup> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

FRISIS / dB

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>P<sub>t</sub></i> — power transmitted (W)<br><i>P<sub>r</sub></i> — power received (W)<br><i>G<sub>t</sub></i> — transmit gain (linear)<br><i>G<sub>r</sub></i> — receive gain (linear)<br><i>S<sub>r</sub></i> = <i>G<sub>t</sub></i> <i>P<sub>t</sub></i> / (4π <i>R</i> <sup>2</sup> ) (W/m <sup>2</sup> )<br><i>P<sub>r</sub></i> = <i>G<sub>t</sub></i> <i>G<sub>r</sub></i> (λ/4π <i>R</i> ) <sup>2</sup> <i>P<sub>t</sub></i><br><i>G</i> = 10 <sup><i>G</i><sub>dB</sub>/10</sup><br>3 dB = ×2, 10 dB = ×10<br><i>P<sub>N</sub></i> = <i>k<sub>B</sub></i> <i>T<sub>sys</sub></i> <i>B</i> (W) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

APERTURE

|                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>A<sub>p</sub></i> — physical area (m <sup>2</sup> )<br>$A_e = \lambda^2 D / (4\pi)$ (m <sup>2</sup> )<br>$A_e \approx A_p$ for aperture<br>$D \approx 4\pi A_p / \lambda^2$ (unitless)<br>β ≈ 0.88λ/ℓ (rect/sq.)<br>β ≈ 1.02λ/ <i>d</i> (circular)<br>2 <i>A<sub>p</sub></i> ⇒ 2 <i>D</i> , β/√2 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

ARRAY

|                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>N</i> — number of elements<br><i>d</i> — spacing (m)<br>γ = <i>k d</i> cos θ<br>$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$<br>$F_a^{\text{max}} = N^2$ at θ = π/2<br>$F_a^{\text{norm}} = F_a / N^2$<br>$F_a \approx 0.88\lambda/(Nd)$ (broadside) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|